

William Makinen

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EDUCATION

Princeton University: Master's in Engineering; GPA: 3.7
M.Eng in Mechanical and Aerospace Engineering

May 2023

Princeton University: Bachelor's of Science in Engineering; GPA: 3.66
B.S.E. in Electrical and Computer Engineering, Computer Science and Robotics Certificates

May 2022

EXPERIENCE

Mytra Inc, Brisbane, CA

June 2024 – Present

- Designing, bringing up, and iterating upon new PCBs and wiring harnesses for Mytra's warehouse automation robots.
- Developing internal tools for PCB part library management and custom PLM integrations and workflows.

Brilliant Smart Home, San Mateo, CA

June 2022 - May 2024

Hardware Team Electrical Engineer

- Improved upon Brilliant's hardware products as core member of the hardware team.
 - Designed, prototyped, and tested PoE variant of Brilliant Control.
 - Debugged critical noise issues within Brilliant Switch product.
- Contributed via circuit design, PCB layout, prototyping, and bring-up.

Brilliant Smart Home, San Mateo, CA

Summer 2022

Hardware Team Intern

- Improved upon Brilliant's hardware products as core member of the hardware team.
 - Researched and implemented doppler radar motion sensor candidates as an alternative to PIR technologies for use in the Brilliant Control.
 - Developed a battery-powered Brilliant Control for use as a demo unit.
- Contributed via circuit design, component selection, prototyping, and PCB layout.

E3D-Online, Oxford, UK

Summer 2021

Intern

- Designed, developed, and tested FDM 3D printing extrusion systems as part of the engineering team.
- Built internal testing rigs and developed accompanying test procedures for company products and prototypes.

OPEX Corporation, Moorestown, NJ

Summer 2019 & Summer 2020

Electrical Engineering Intern

- 2020: Worked in Incoming and Scanning Division on variety of circuit design and embedded systems projects.
- 2019: Worked in Warehouse Automation Division to streamline and automate final QC process for SSXL iBots.

U.S. Naval Research Laboratory, Washington, DC

Summer 2017 & Summer 2018

Student Researcher

- 2018: Experimented with the 3D printing of technical ceramics as an alternative fabrication method to previous year's work.
- 2017: Worked in Optical Sciences division to construct a binder-jetter 3D printer capable of forming technical ceramics into complex shapes.

Intel International Science and Engineering Fair (ISEF), Pittsburgh, PA and Los Angeles, CA

Award Winner

May 2017

- Developed fully autonomous system for detecting and correcting errors during the 3D printing process. Received 4th place in Engineering and Mechanics category at international fair.
- **US Patent Application Granted:** US Patent No. 11,084,091.

Participant

March 2016

- Built autonomous, computer vision-controlled object collection robot to seek out and collect designated objects in a room. Intended to help the disabled keep homes clean and safe. Placed 2nd at regional fair.

Finalist

May 2015

- Constructed a robotic hand designed to assist with recovery process of temporary forearm injuries (broken radii, etc.). Grand prize at Fairfax County regional fair, finalist at international fair.

Student Researcher

- Worked with other research engineers on real-time 3D printer characterization research to detect print errors in real time. Work expanded upon for 2017 Intel ISEF project.

ACTIVITIES

West Valley Track Club

12-15 hours per week

Social Chair

- Organize team social activities, help new members feel at home in the club, and assist with recruitment.
- Compete in track, road, and cross country races, attend workouts/practices, and volunteer at club events/races.

SKILLS

Python, C/C++, Java, MATLAB, Linux Systems, Circuit Design and Simulation, PCB Design and Layout, Verilog and FPGA, CAD (SolidWorks, Inventor/Fusion360), G-Code and CNC Devices, Oscilloscopes, DMMs, Signal Generators, Workshop Tools and General Making.